

An Overlay for Social Networks based on Directed Acyclic Graphs on the Edge

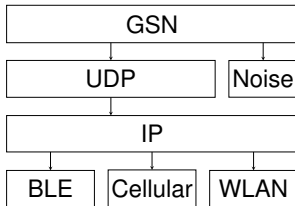
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Chair of Connected Mobility
School of Computation, Information, and Technology
Technical University of Munich

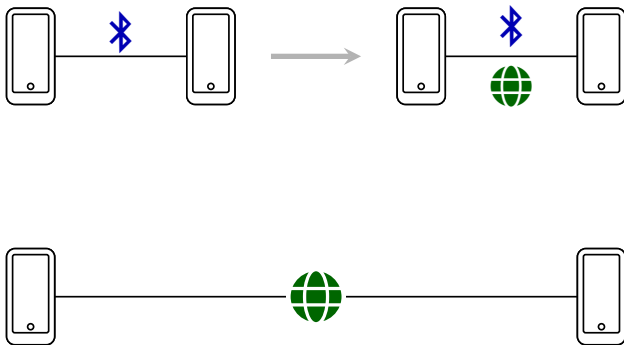


Architecture



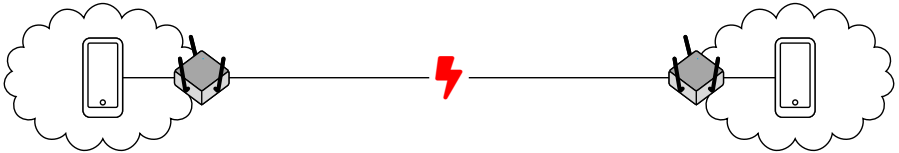
- Connect two peer devices according to priority and proximity pair
- Dual-stack network comms: BLE first, Cellular (5G/LTE) and WLAN second
- In close range, use BLE for signaling to establish UDP connection
- ICE-similar stack for address reflection, hole-punching using friends

Dual-stack comms

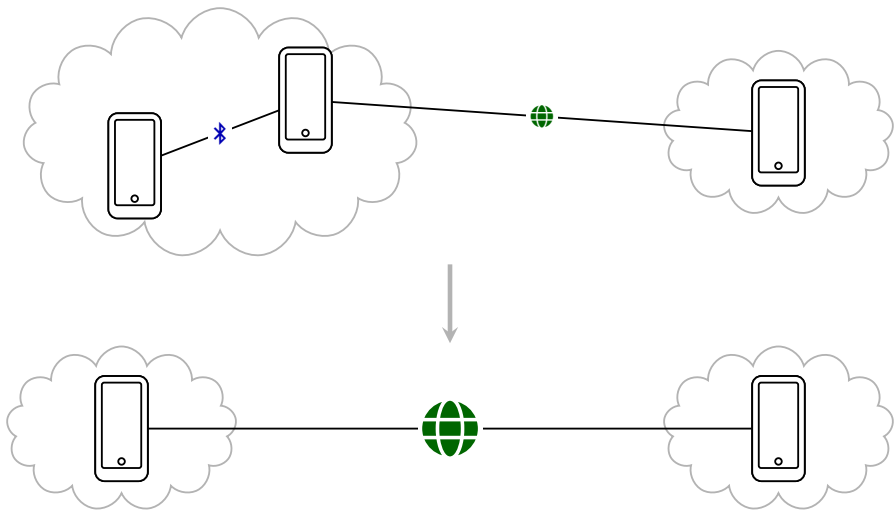


- BLE as signaling channel
- Advertise multiple addresses
- UDP for hole-punching
- UDX for long-range communication

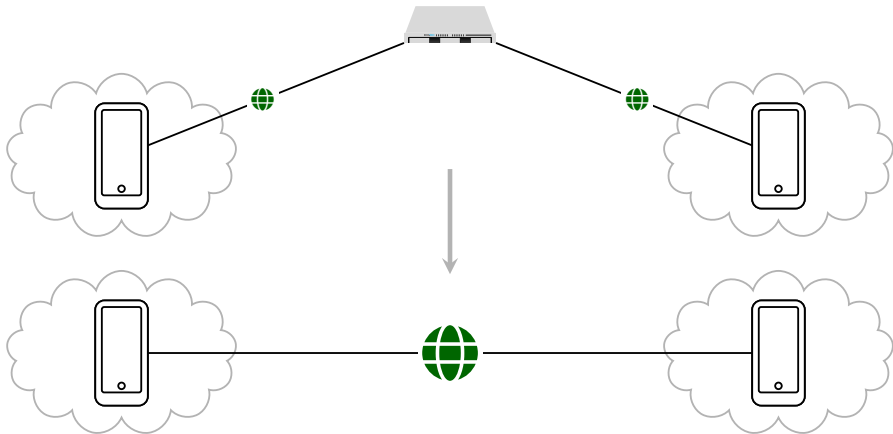
NAT



Well-connected friends



Rendezvous Server



Since mid-April: two close projects

- Built the transport around cryptographic identity: Ed25519 public keys as peer identity and signed packets as the basic trust object.
- Stabilized a shared substrate: BLE discovery, friend relations, Noise sessions, fragmentation, receipts, and Redux-backed transport state.
- Worked through NAT traversal: public address discovery, hole-punching, well-connected friends, and rendezvous-style coordination.
- Separated the work into two related regimes: **city-scale BLE relation** and **Internet-scale UDX relation**.
- Identified the common security need: relays and rendezvous helpers may route metadata, but only the final recipient should read content.
- Framed the master thesis around BLE mobility and local relay evaluation, while preserving UDX as a comparison and Keidar–Shapiro research track.
- Connected both tracks to opportunistic networking, grassroots systems, decentralization critiques, and blocklace-style accountability.

Mobility model: city-scale BLE relay

Goal

- Test if BLE relay can replace UDP locally.
- Compare direct, friend, and open relay (simulated).
- Report delivery, latency, overhead, and abuse surface.

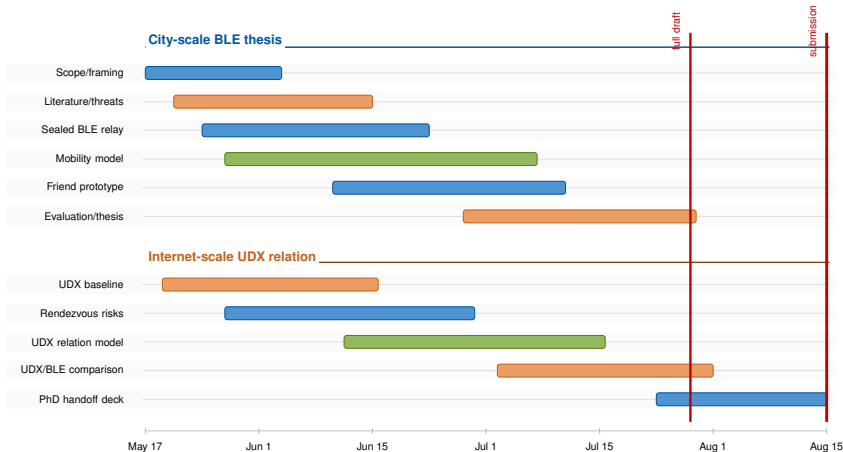
What we track

- Encounters: pseudonym pair, start/end, duration, inter-contact, RSSI band.
- Delivery: created, relayed, ACKed, hop count, TTL expiry, size, failure.
- Cost: relay bytes, duplicates, queues, scan/advertise time, battery proxy.

How

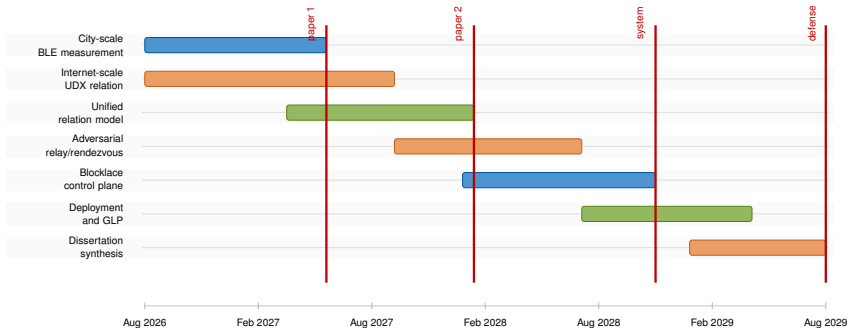
- Encrypted local trace log.
- Upload only by explicit opt-in.
- Replay traces in simulator calibrated by BLE contacts.
- No content; coarse location only by separate opt-in.

Master gantt: thesis plus UDX track



- Thesis core: BLE mobility and secure local relay; Keidar–Shapiro thread: keep UDX as an Internet-scale relation model and comparison.

PhD runway: two relation systems



- Thesis-to-PhD bridge: compare what grassroots relations mean under local physical mobility and under Internet-scale UDX connectivity.